

Vancouver Crime Pattern Analysis: Available Datasets & Interactive Dashboard Development Plan

Executive Summary

This document outlines available datasets for analyzing crime patterns in Vancouver using persistent homology and environmental correlation analysis. It provides a comprehensive plan for creating an interactive dashboard to understand the relationship between urban infrastructure, environmental factors, and crime patterns.

1. AVAILABLE DATASETS FROM OPEN DATA PORTALS

1.1 CRIME DATA

Vancouver Police Department (VPD) GeoDASH

- **Source:** <https://geodash.vpd.ca/opendata/>
- **Data Available:**
 - Crime incidents by type, location (hundred block level), date/time
 - Hundred block locations (coordinates offset for privacy)
 - Historical data from 2003-present
 - Crime types: theft from vehicle, break & enter, mischief, assault, theft of vehicle, etc.
- **Format:** CSV, Shapefile, API access
- **Update Frequency:** Monthly/Quarterly
- **Note:** Coordinates are intentionally offset for privacy protection

City of Vancouver Open Data Portal - Crime Indicators

- **Dataset:** Indicator Data (Crime section)
 - **Link:** <https://opendata.vancouver.ca/explore/dataset/indicator-data/>
 - **Contains:** Violent and property crime incidents by quarter
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1.2 TRANSIT & TRANSPORTATION DATA

Rapid Transit Stations

- **Dataset:** Rapid transit stations
- **Link:** <https://opendata.vancouver.ca/explore/dataset/rapid-transit-stations/>

- **Contains:** SkyTrain station locations (Expo, Millennium, Canada Lines)
- **Format:** Point features with coordinates

TransLink GTFS Data

- **Source:** <https://www.translink.ca/about-us/doing-business-with-translink/app-developer-resources/gtfs/gtfs-data>
- **Contains:**
 - Bus stop locations (8,500+ stops)
 - Transit routes and patterns
 - Station locations
 - Real-time transit data (GTFS Realtime)
- **Format:** GTFS feed, GeoJSON, Shapefile
- **API Available:** Yes (requires API key)

Traffic Signals

- **Dataset:** Traffic signals
 - **Link:** <https://opendata.vancouver.ca/explore/dataset/traffic-signals/>
 - **Contains:** Traffic signal locations at intersections
 - **Update:** Weekly
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1.3 STREET LIGHTING

Street Lighting Poles

- **Dataset:** Street lighting poles
- **Link:** <https://opendata.vancouver.ca/explore/dataset/street-lighting-poles/>
- **Contains:**
 - Pole locations (survey accuracy)
 - Street lighting network nodes
- **Format:** Point features with coordinates
- **Update:** Weekly

Street Lighting Conduits

- **Dataset:** Street lighting conduits
 - **Link:** <https://opendata.vancouver.ca/explore/dataset/street-lighting-conduits/>
 - **Contains:** Street light conduit/duct network
 - **Format:** Line features
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1.4 PROPERTY & ZONING DATA

Property Tax Reports

- **Dataset:** Property tax report (2006-present)
- **Link:** <https://opendata.vancouver.ca/explore/dataset/property-tax-report/>
- **Contains:**
 - Property assessments (land value, improvement value)
 - Tax levy information
 - Property coordinates
 - Zoning information
 - Year built
 - Neighbourhood codes
- **Format:** CSV, API
- **Update:** Weekly (current year)

Zoning Districts and Labels

- **Dataset:** Zoning districts and labels
 - **Link:** <https://opendata.vancouver.ca/explore/dataset/zoning-districts-and-labels/>
 - **Contains:**
 - Zoning polygons
 - Zoning categories (residential, commercial, industrial, etc.)
 - Development regulations
 - **Format:** Polygon features
 - **Update:** Weekly
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1.5 BUILDING & INFRASTRUCTURE

Building Footprints

- **Datasets Available:**
 - Building footprints 2015 (most recent)
 - Building footprints 2009
 - Building footprints 1999 (downtown only)
- **Link:** <https://opendata.vancouver.ca/explore/dataset/building-footprints-2015/>
- **Contains:**
 - 2D building outlines
 - Building geometries at parcel level
- **Format:** Polygon features
- **Note:** Static datasets, not updated

Parks Data

- **Dataset:** Parks & Parks facilities
 - **Link:** <https://opendata.vancouver.ca/explore/dataset/parks/>
 - **Contains:**
 - Park boundaries (220+ parks)
 - Park facilities
 - Green space locations
 - **Format:** Polygon and point features
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1.6 BUSINESS & COMMERCIAL DATA

Business Licences

- **Dataset:** Business licences
- **Link:** <https://opendata.vancouver.ca/explore/dataset/business-licences/>
- **Contains:**
 - Business name, type, address
 - Coordinates for location mapping
 - Business categories (100+ types including bars, restaurants, retail, gas stations)

- License status (active/inactive)
 - Historical data from 1997-present
 - **Format:** CSV, API
 - **Update:** Daily
 - **Use Cases:**
 - Identify commercial establishments by type
 - Map bars, liquor establishments, gas stations
 - Analyze commercial density
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1.7 CENSUS & DEMOGRAPHICS DATA

Statistics Canada Census Data

- **Source:** Statistics Canada Census Profile 2021
- **Link:** <https://www12.statcan.gc.ca/census-recensement/2021/dp-pd/prof/>
- **Contains:**
 - Population demographics
 - Income levels
 - Education
 - Housing characteristics
 - Language
 - Employment data

Census Local Area Profiles

- **Dataset:** Census local area profiles 2016
 - **Link:** <https://opendata.vancouver.ca/explore/dataset/census-local-area-profiles-2016/>
 - **Contains:** Census data for Vancouver's 22 local planning areas
 - **Variables:** Population, age, household types, income, education, employment, dwelling types
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1.8 ADDITIONAL INFRASTRUCTURE DATA

Street Network

- **Available:** Street centerlines, public/non-public streets

- **Use:** Calculate block lengths, street network analysis

Neighbourhood Boundaries

- **Available:** Local area boundaries (22 areas)
- **Use:** Spatial aggregation, neighborhood-level analysis

Gas Stations

- **Source:** Business licenses dataset (filter by business type: "Service Station" or "Gas Station")
 - **Use:** Map potential observation points
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2. ADDITIONAL DATA SOURCES FROM OFFICIAL WEBSITES

2.1 Statistics Canada

- **CensusMapper:** <https://censusmapper.ca/>
 - Interactive census data visualization
 - Dissemination area level data
 - Custom geographic queries

2.2 BC Assessment

- **Source:** <https://www.bcassessment.ca/>
- **Contains:** Property assessment values, building characteristics

2.3 Metro Vancouver Open Data

- **Source:** <https://open-data-portal-metrovancouver.hub.arcgis.com/>
- **Contains:** Regional datasets beyond city boundaries

2.4 ICBC (Insurance Corporation of BC)

- **Traffic collision data:** May require special request
 - **Intersection safety ratings**
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3. DATA GAPS & SOLUTIONS

Data NOT Available in Open Portals (Need Alternative Approaches):

1. Red Light Cameras:

- Contact ICBC directly
- May be partially available in traffic signal dataset

2. Security Cameras:

- Not publicly available (privacy concerns)
- Proxy: Commercial establishment density

3. Street Light Illumination Range:

- Not directly available
- **Solution:** Use standard lighting specifications (e.g., LED street lights typically illuminate 20-30m radius)
- Calculate coverage using buffer analysis

4. Detailed Building Heights/3D Models:

- Building footprints available but not heights
- **Solution:**
 - Use LiDAR data if available
 - Estimate from building age and zoning
 - Request from City of Vancouver Planning Department

5. Pedestrian Traffic Volume:

- Limited availability
- Proxy: Commercial density, transit ridership data

4. INTERACTIVE DASHBOARD DEVELOPMENT STRATEGY

4.1 Dashboard Architecture

Technology Stack Recommendations:

Option 1: Web-Based Dashboard (Recommended)

- **Frontend:** React.js with Deck.gl or Mapbox GL JS
- **Backend:** Python (Flask/FastAPI) or Node.js

- **Database:** PostGIS (PostgreSQL with spatial extension)
- **Visualization:**
 - D3.js for charts
 - Deck.gl for 3D visualizations
 - Plotly Dash for interactive plots

Option 2: Desktop Application

- **QGIS** with Python plugins
- **ArcGIS Pro** with custom toolboxes
- **Tableau** or **Power BI** for business intelligence

Option 3: Python-Based

- **Streamlit** or **Dash** for rapid prototyping
- **Folium** for mapping
- **Plotly** for interactive charts

4.2 Dashboard Core Features

Layer 1: Base Map

- **Street network** (from OSM or City data)
- **Neighborhood boundaries**
- **Building footprints**
- **Parks and green spaces**

Layer 2: Crime Data Visualization

- **Heatmaps:** Crime density by type and time period
- **Point maps:** Individual incidents (with privacy offset)
- **Temporal animation:** Crime patterns over time
- **Crime type filtering:** Select specific crime categories

Layer 3: Environmental Features (Toggleable)

1. Transit Infrastructure

- SkyTrain stations with radius buffers (100m, 200m, 500m, 1km)

- Bus stops with coverage areas
- Transit routes
- Interactive: Select station to see crime within X meters

2. Street Lighting

- Street light pole locations
- Illumination coverage zones (estimated 25m radius per light)
- **3D Visualization:** Represent lights as yellow/orange spheres with glow effect
- Dark zones (areas with low lighting density)

3. Commercial Establishments

- All businesses (from license data)
- Specific types: Bars, liquor stores, gas stations, convenience stores
- Density heatmaps by business type

4. Property Characteristics

- Property values (choropleth map)
- Property tax levels
- Building age
- Zoning categories

5. Traffic Infrastructure

- Traffic signals
- Major roads/highways
- Traffic volume (if available)

Layer 4: Block-Level Analysis

- **Block boundaries:** Custom-created from street network
- **Block characteristics:**
 - Block length (perimeter)
 - Block area/compactness ratio
 - Number of buildings per block
 - Crime count per block
 - Average property value
 - Zoning diversity index

Layer 5: Similarity Analysis

- **Color-coded blocks** by similarity threshold (10%, 20%, 30%+)
 - **Similarity metrics:**
 - Property value similarity
 - Zoning similarity
 - Commercial density similarity
 - Crime pattern similarity
 - **Clustering visualization:** Blocks grouped by similar characteristics
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4.3 Interactive Controls

Filters & Selectors:

1. **Time Range:** Date slider for temporal analysis
2. **Crime Type:** Multi-select dropdown
3. **Neighborhood:** Select specific areas
4. **Layer Toggle:** Show/hide each data layer
5. **Similarity Threshold:** Slider (0-100%)
6. **Distance from Transit:** Radius selector (100m - 2km)

Analysis Tools:

1. **Buffer Analysis:** Click transit station → see crimes within radius
 2. **Correlation Matrix:** Crime vs. environmental variables
 3. **Statistical Summary:** For selected area/block
 4. **Export:** Download filtered data as CSV/GeoJSON
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4.4 3D Visualization Approach

Creating 3D Vancouver

Approach 1: Web-Based 3D (Recommended for Accessibility)

Libraries:

- **Deck.gl** (React/JavaScript)

- Excellent for large datasets
- GPU-accelerated rendering
- Built-in 3D building layer
- **Cesium.js**
 - True 3D globe rendering
 - Time-dynamic visualization
 - Handles terrain elevation

Implementation:

```
javascript
```

```
// Example with Deck.gl
import {HexagonLayer, GeoJsonLayer, ScatterplotLayer} from '@deck.gl/layers';

const layers = [
  // Building footprints as 3D extrusions
  new GeoJsonLayer({
    id: 'buildings',
    data: buildingData,
    extruded: true,
    getElevation: f => estimateHeight(f.properties.year_built, f.properties.zone),
    getFillColor: [74, 80, 87]
  }),

  // Street lights as glowing spheres
  new ScatterplotLayer({
    id: 'street-lights',
    data: streetLightData,
    getPosition: d => d.coordinates,
    getRadius: 25, // 25m illumination radius
    getFillColor: [255, 215, 0, 100], // Yellow glow
    opacity: 0.6
  }),

  // Crime incidents as elevated points
  new ScatterplotLayer({
    id: 'crimes',
    data: crimeData,
    getPosition: d => d.coordinates,
    getRadius: 10,
    getFillColor: d => getCrimeColor(d.type),
    elevationScale: 50
  })
];
```

Approach 2: Desktop 3D

ArcGIS Pro / QGIS 3D:

- Import all datasets as layers
- Use 2.5D or 3D visualization
- Apply height extrusion for buildings
- Create custom symbology for street lights (glow effect)

Approach 3: Game Engine (Advanced)

Unity or Unreal Engine:

- Import geospatial data
 - Create realistic 3D city model
 - Add lighting simulation
 - Interactive crime visualization
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4.5 Street Light Visualization Strategy

Since exact illumination data isn't available:

1. Create Light Coverage Model:

- Each street light = point source
- Apply standard illumination radius (20-30m)
- Use graduated symbology: bright yellow center → fading orange → dark

2. 2D Representation:

- Heat map showing illuminated areas
- Dark areas (low light density) in dark blue/black
- Color scale: Dark blue (0 lights) → Yellow (high density)

3. 3D Representation:

- Street light poles as vertical lines/cylinders
- Glowing spheres at pole tops
- Light cone projecting downward (30° spread angle)
- Ambient light effect on surrounding ground
- **Visual Impact:** Bright spots create "constellation" effect showing lighting network

4. Correlation with Crime:

- Overlay crime points on lighting layer
 - Calculate: % crimes in well-lit vs. poorly-lit areas
 - Statistical test: Chi-square or spatial autocorrelation
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4.6 Persistent Homology Integration

Data Preparation for TDA (Topological Data Analysis):

1. Create Point Cloud:

- Crime locations as primary points
- Environmental features as weighted points:
 - Street lights (positive weight - safety)
 - Dark zones (negative weight)
 - Transit stations
 - Commercial establishments
 - Property values

2. Feature Vectorization: For each spatial unit (block/grid cell):

- Crime count by type
- Street light density
- Transit proximity score
- Commercial density
- Property value metrics
- Zoning diversity

3. Persistent Homology Analysis:

- Use **GUDHI** (Python) or **TDA** (R) libraries
- Compute persistence diagrams
- Identify topological features:
 - Connected components (H_0): Crime clusters
 - Loops (H_1): Circular patterns, "voids" in crime distribution
 - Voids (H_2): 3D structures in multivariate space

4. Visualization:

- Persistence diagrams
- Barcodes showing feature lifespans
- Map topological features back to geographic space

Example Workflow:

```
python
```

```
import gudhi
import numpy as np

# Create feature matrix for each block
# features = [crime_count, light_density, transit_dist, property_value, ...]
feature_matrix = create_feature_matrix(blocks)

# Compute Rips complex
rips_complex = gudhi.RipsComplex(points=feature_matrix, max_edge_length=2.0)
simplex_tree = rips_complex.create_simplex_tree(max_dimension=2)

# Compute persistence
persistence = simplex_tree.persistence()

# Visualize
gudhi.plot_persistence_diagram(persistence)
```

4.7 Dashboard Development Phases

Phase 1: Data Collection & Preprocessing (Weeks 1-2)

- Download all datasets from open portals
- Clean and standardize formats
- Create unified spatial reference system (EPSG:4326 or EPSG:3157)
- Set up PostGIS database
- Import all data layers

Phase 2: Exploratory Analysis (Weeks 3-4)

- Basic spatial analysis in QGIS/ArcGIS
- Statistical correlations
- Identify patterns and relationships
- Create block-level aggregations

Phase 3: Dashboard Framework (Weeks 5-6)

- Set up web application structure
- Create base map with layer toggles
- Implement basic filtering

- Add time slider for temporal analysis

Phase 4: Advanced Features (Weeks 7-8)

- 3D visualization implementation
- Street light coverage modeling
- Transit proximity analysis
- Block similarity calculations

Phase 5: Persistent Homology Integration (Weeks 9-10)

- Feature engineering
- TDA computation
- Visualization of topological features
- Interactive persistence diagram exploration

Phase 6: Testing & Refinement (Weeks 11-12)

- User testing
 - Performance optimization
 - Documentation
 - Deployment
-

5. SPECIFIC ANALYSIS RECOMMENDATIONS

5.1 Transit Proximity Analysis

Research Questions:

- How does crime rate change with distance from SkyTrain stations?
- Is there a threshold distance where crime drops significantly?
- Do different crime types show different patterns?

Methods:

1. Buffer Analysis:

- Create buffers around each station (50m, 100m, 200m, 500m, 1km, 2km)
- Count crimes in each buffer zone

- Calculate crime rate per area

2. Distance Decay Analysis:

- For each crime, calculate nearest SkyTrain distance
- Plot crime density vs. distance
- Fit decay curve (exponential, polynomial)

3. Statistical Testing:

- Compare crime rates: near-transit vs. far-transit
- Control for confounding variables (property values, commercial density)
- Regression analysis

5.2 Block Characteristics Analysis

Metrics to Calculate:

1. Geometric:

- Block area (m²)
- Block perimeter (m)
- Compactness ratio: $4\pi \times \text{Area} / \text{Perimeter}^2$
- Number of buildings per block
- Building density (buildings per hectare)

2. Crime Metrics:

- Crimes per block
- Crimes per building
- Crimes per unit area
- Crime diversity (Shannon entropy of crime types)

3. Urban Network Analysis (using UNA tool):

- Reach: Number of destinations reachable within distance
- Betweenness centrality: How "between" other places a block is
- Closeness centrality: Average distance to all other blocks
- Straightness: Ratio of Euclidean to network distance

5.3 Similarity Measure Implementation

Approach:

1. Feature Normalization:

- Standardize all variables (z-scores)
- Handle missing data

2. **Similarity Calculation:**

- Euclidean distance in feature space
- Convert to similarity: $\text{similarity} = 1 / (1 + \text{distance})$
- Or use cosine similarity

3. **Thresholding:**

- Define similarity thresholds ($0.7 = 70\%$ similar)
- Group blocks exceeding threshold
- Visualize on map with color coding

4. **Variables for Similarity:**

- Property value (median, mean)
- Property tax (average)
- Zoning composition (% residential, commercial, industrial)
- Building age distribution
- Crime rate
- Commercial density
- Transit accessibility

6. SOFTWARE & TOOLS RECOMMENDATIONS

6.1 Spatial Analysis

- **QGIS** (free): General GIS operations, visualization
- **ArcGIS Pro** (paid): Advanced spatial analysis, professional outputs
- **PostGIS**: Spatial database queries
- **GeoPandas** (Python): Programmatic spatial analysis

6.2 Network Analysis

- **Urban Network Analysis (UNA):**
 - Grasshopper plugin for Rhino
 - City Form Lab tool
 - Calculate reach, betweenness, closeness

- **OSMnx** (Python):
 - Download street networks from OpenStreetMap
 - Network analysis
 - Shortest paths, centrality measures

6.3 Statistical Analysis

- **R**: Statistical modeling, spatial statistics
- **Python** (pandas, scipy, scikit-learn): Data analysis, ML
- **SPSS** or **Stata**: Traditional statistical analysis

6.4 Visualization

- **Tableau**: Interactive dashboards (no coding)
- **Power BI**: Business intelligence dashboards
- **Plotly Dash** (Python): Programmable dashboards
- **D3.js** (JavaScript): Custom interactive visualizations
- **Kepler.gl**: Geospatial data visualization (open source)

6.5 Persistent Homology

- **GUDHI** (Python/C++): Geometry Understanding in Higher Dimensions
- **Ripser** (Python): Fast persistence computation
- **TDA** (R package): Topological data analysis
- **giotto-tda** (Python): Machine learning with TDA

7. EXAMPLE DASHBOARD FEATURES

Feature 1: "Crime Around Transit" Tool

User Action: Click on any SkyTrain station **Dashboard Response:**

- Highlight station
- Draw concentric circles (100m, 500m, 1km)
- Show crime counts in each zone
- Display bar chart of crime types
- Calculate crime density (crimes per km²)

- Compare to city average

Feature 2: "Street Light Analysis"

User Action: Select neighborhood **Dashboard Response:**

- Show 3D view with street lights as glowing points
- Highlight dark zones (low light density)
- Overlay crime points
- Calculate:
 - % crimes in well-lit areas
 - % crimes in dark zones
- Statistical significance test result

Feature 3: "Block Similarity Explorer"

User Action: Adjust similarity threshold slider **Dashboard Response:**

- Re-color blocks based on similarity groups
- Show statistics for each group
- Allow selection of representative block
- Display detailed comparison of similar blocks

Feature 4: "Persistent Homology Viewer"

User Action: Select time period and features **Dashboard Response:**

- Compute and display persistence diagram
- Show longest-lived topological features
- Map features back to geographic space
- Highlight significant crime clusters and voids

8. DATA PROCESSING WORKFLOW

Step 1: Data Download & Organization

```
project/  
├── data/  
│   ├── raw/
```

```
| | |— crime/
| | |— transit/
| | |— lighting/
| | |— property/
| | |— business/
| | |— census/
| |— processed/
| |— outputs/
|— scripts/
| |— 01_download_data.py
| |— 02_clean_data.py
| |— 03_spatial_join.py
| |— 04_feature_engineering.py
| |— 05_analysis.py
|— dashboard/
| |— app.py
| |— layouts/
| |— callbacks/
|— docs/
```

Step 2: Data Cleaning & Standardization

- Ensure consistent CRS (Coordinate Reference System)
- Handle missing coordinates
- Remove duplicates
- Standardize date formats
- Create unique IDs for all features

Step 3: Spatial Integration

- Spatial join crime to blocks
- Calculate distance from each crime to nearest transit
- Assign each crime to nearest street light
- Link businesses to blocks
- Aggregate census data to blocks

Step 4: Feature Engineering

- Create block-level summary statistics
- Calculate distance metrics

- Compute density measures
- Derive similarity scores
- Generate temporal features (day of week, hour, season)

Step 5: Dashboard Deployment

- Local deployment (for development)
 - Cloud deployment (Heroku, AWS, Azure)
 - Continuous updates from data sources
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9. EXPECTED INSIGHTS FROM ANALYSIS

Correlation Patterns to Investigate:

1. Transit-Crime Relationship:

- Hypothesis: Crime peaks within 200m of SkyTrain stations
- Expected: Exponential decay with distance
- Crime types: Theft more common near transit; violent crime less affected

2. Street Lighting Effect:

- Hypothesis: Well-lit areas have lower crime rates
- Expected: 20-30% reduction in crime in well-lit areas
- Type dependency: Property crimes more affected than violent crimes

3. Commercial Density:

- Hypothesis: High commercial density = higher crime (more opportunities)
- Expected: Positive correlation, especially for theft and property crime
- Specific establishments: Bars/liquor stores correlate with assault

4. Property Values:

- Hypothesis: Complex relationship - very high and very low values both associate with crime
- Expected: U-shaped curve
- Mechanism: Low value = economic distress; High value = attractive targets

5. Block Compactness:

- Hypothesis: More compact blocks = lower crime (better natural surveillance)
- Expected: Negative correlation with compactness ratio

6. Topological Features (from Persistent Homology):

- Crime clusters form persistent connected components
 - "Voids" (low-crime areas) correspond to well-lit residential zones
 - Loop structures may indicate crime displacement patterns
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10. NEXT STEPS & RECOMMENDATIONS

Immediate Actions:

1. Download Core Datasets:

- Crime data (VPD GeoDASH)
- Transit stations and GTFS
- Street lighting poles
- Property tax reports
- Business licenses
- Zoning data

2. Set Up Development Environment:

- Install Python with spatial libraries (geopandas, shapely, folium)
- Install QGIS for initial exploration
- Set up PostGIS database

3. Conduct Initial EDA:

- Map crime patterns
- Visualize transit network
- Explore street light distribution
- Identify data quality issues

Medium-term Actions:

4. Develop Block Segmentation:

- Use street network to define blocks
- Calculate block metrics
- Aggregate data to block level

5. Build Similarity Framework:

- Select variables for similarity
- Compute pairwise similarities

- Test different thresholds
- Validate groupings

6. Create Prototype Dashboard:

- Start with 2D version
- Add basic filtering
- Implement key visualizations

Long-term Actions:

7. Implement 3D Visualization:

- Add street light glow effects
- Extrude buildings
- Create immersive crime view

8. Integrate Persistent Homology:

- Feature engineering for TDA
- Compute persistence
- Interpret topological features
- Link back to policy recommendations

9. Validate & Publish:

- Statistical validation of findings
- Academic paper on methodology
- Public-facing dashboard for community

11. POTENTIAL CHALLENGES & SOLUTIONS

Challenge 1: Privacy & Ethics

Issue: Crime data offset for privacy; showing exact locations may violate privacy **Solution:**

- Use hundred-block aggregation
- Show density/heat maps rather than points
- Never link to individual identities

Challenge 2: Data Volume

Issue: Large datasets may slow dashboard **Solution:**

- Pre-compute aggregations
- Use tile servers for mapping
- Implement lazy loading
- Add progress indicators

Challenge 3: Missing Data

Issue: Some addresses lack coordinates; building heights unavailable **Solution:**

- Use geocoding services for missing coordinates
- Estimate building heights from year built and zoning
- Document assumptions and limitations

Challenge 4: Temporal Dynamics

Issue: Static snapshots may miss temporal patterns **Solution:**

- Implement time slider
- Show animated heat maps
- Compare different time periods
- Account for seasonal variations

Challenge 5: Causation vs. Correlation

Issue: Correlation doesn't imply causation **Solution:**

- Use causal inference methods (instrumental variables, DID)
- Control for confounders
- Be explicit about limitations
- Frame findings as associations, not causes

12. SUMMARY OF KEY DATASETS & TOOLS

Essential Datasets (Must Have):

✓ Crime data (VPD GeoDASH)

- ✓ Transit stations (City Open Data)
- ✓ Street lighting poles (City Open Data)
- ✓ Property tax/values (City Open Data)
- ✓ Business licenses (City Open Data)
- ✓ Zoning districts (City Open Data)

Supplementary Datasets (Nice to Have):

- Census demographics (Statistics Canada)
- Building footprints (City Open Data)
- Parks data (City Open Data)
- Traffic signals (City Open Data)
- Street network (OSM or City data)

Core Tools:

- **Python:** Data processing, analysis, TDA
 - **QGIS/ArcGIS:** Spatial analysis, visualization
 - **Deck.gl or Mapbox:** 3D web visualization
 - **GUDHI/Ripser:** Persistent homology computation
 - **Streamlit/Dash:** Interactive dashboard framework
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CONCLUSION

This comprehensive plan provides a roadmap for:

1. Accessing all necessary datasets from open portals
2. Building an interactive dashboard with multiple layers
3. Implementing advanced spatial analysis including persistent homology
4. Creating 3D visualizations to understand Vancouver's crime-environment relationships

The combination of traditional GIS analysis with topological data analysis (persistent homology) represents a novel approach to understanding crime patterns, potentially revealing hidden spatial structures that conventional methods might miss.

Key Strength: All primary datasets are freely available, updated regularly, and provided by authoritative sources (City of Vancouver, VPD, TransLink, Statistics Canada).

Innovation: Applying persistent homology to crime-environment correlation is a cutting-edge approach that could provide new insights for urban planning and crime prevention.

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Data sources verified as of: February 2026

Next update recommended: Quarterly